

Drug loading techniques for exosome-based drug delivery systems

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Abstract

Exosomes are bilayer membrane-coated extracellular vesicles measuring between 40 and 100 nm in diameter. As a natural carrier, exosomes have the advantages of low immunogenicity, high stability in blood, and direct delivery of drugs to cells. Exosomes can be transported between cells and thus are conducive to the exchange of substances and information between cells. They change the functional state of recipient cells by loading exogenous drugs (e.g., small-molecule drugs, transmembrane proteins, and nucleic acid drugs). The key to using exosomes as drug carriers is the effective loading of exogenous drugs into exosomes; however, this task poses a challenge in studying the functionalization of exosomes as drug carriers. Currently, sonication, electroporation, transfection, incubation, extrusion, saponin-assisted loading, transgenesis, freeze-thaw cycles, thermal shock, pH gradient method, and hypotonic dialysis have been applied to load these drugs into exosomes. This review aims to provide an overview of the advantages and disadvantages of various drug loading technologies for exosomes.